



ORIENT YOUR ATTENTION IN TIME

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INTRODUCTION

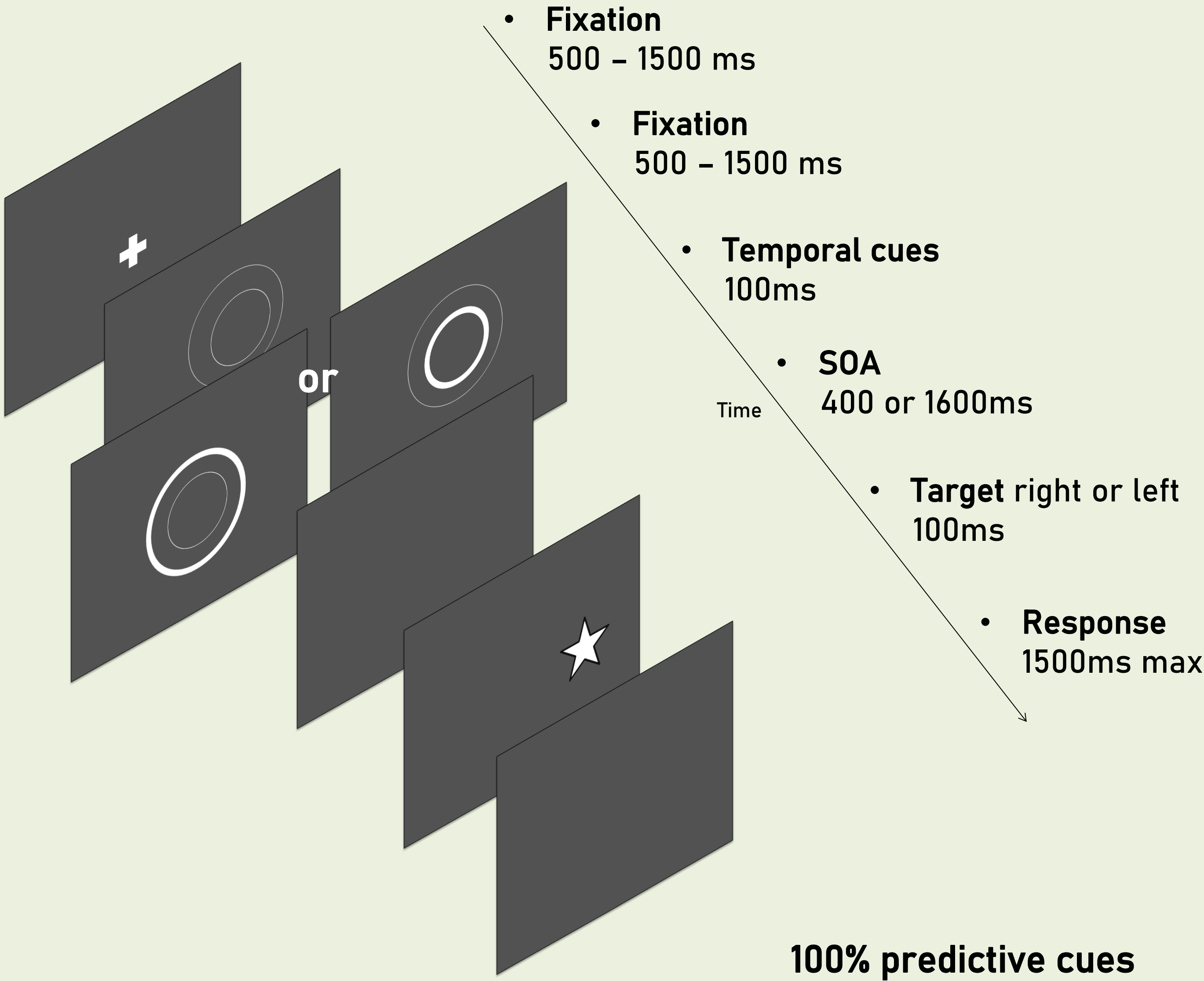
- Limited processing capacity, Humans have to select the most relevant information for adaptative behaviour . Information prioritization can be given by orienting attention at a particular moment in time.
- Four ways to orient attention in time, depending on the nature of cues : endogenous temporal orienting, rhythmic cueing, sequential effects and foreperiod effect (Nobre & van Ede, 2018; Capizzi & Correa, 2018).
- OBJECTIVES: To design a paradigm to assess endogenous, exogenous and rhythmic modes of orienting

ENDOGENOUS PROCEDURE

ENDOGENOUS TEMPORAL ORIENTING

Manipulated on a trial basis

(Coull & Nobre, 1998)

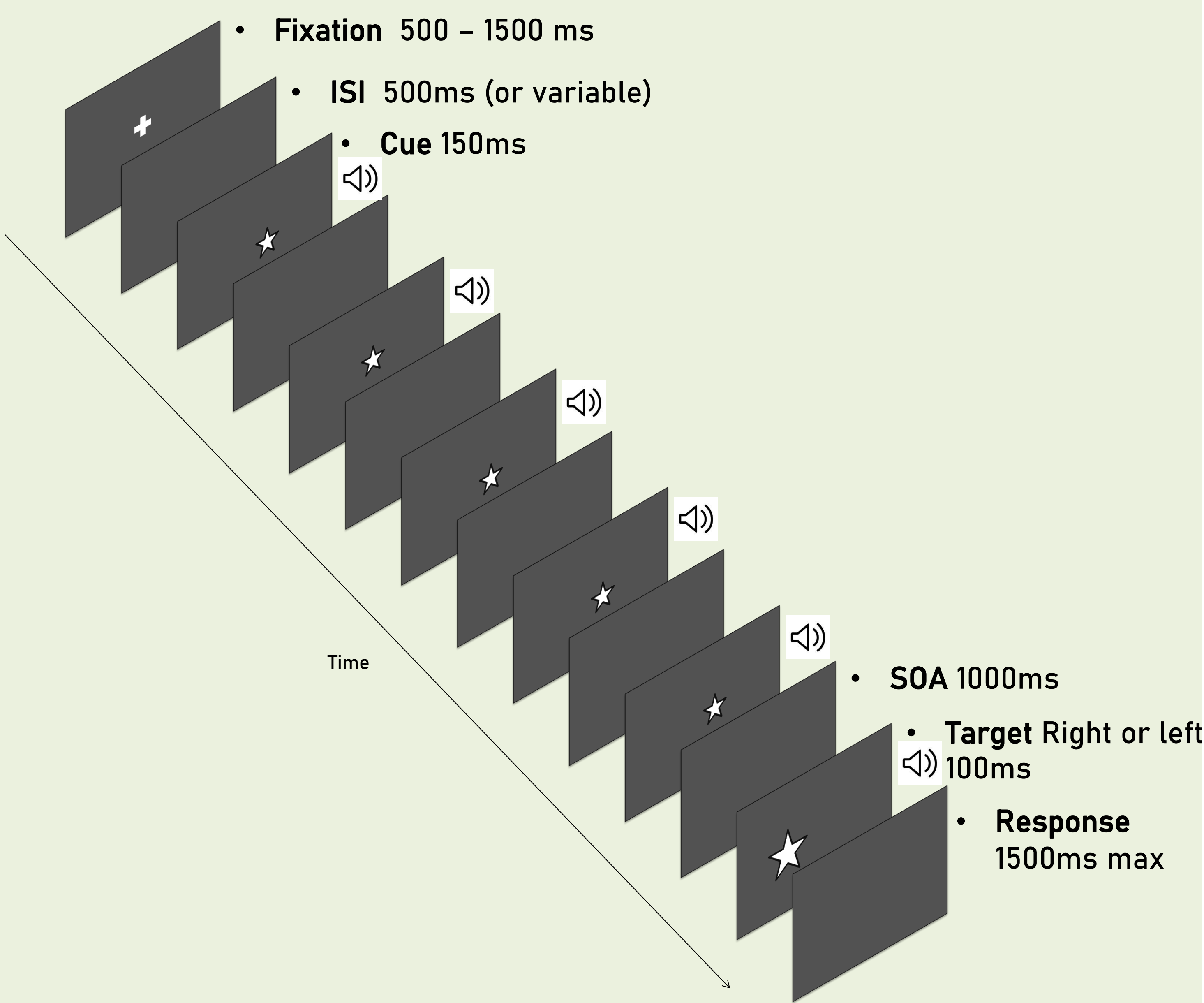


RHYTHMIC PROCEDURE

RHYTHMIC ORIENTING

Manipulated with isochronous and asynchronous rhythms

(Sanabria *et al.*, 2011; Cutanda *et al.*, 2015)



FOREPERIOD EFFECT

Manipulated through cue – target intervals (400, 800, 1200 or 1600ms)

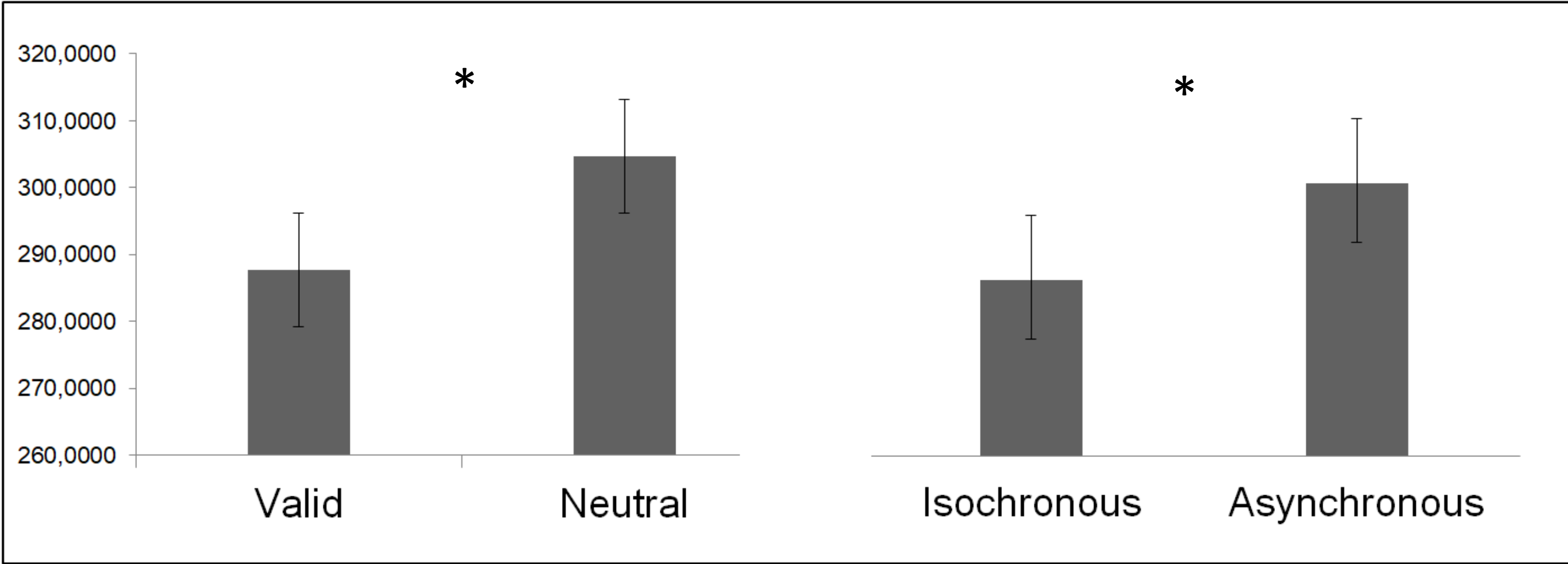
SEQUENTIAL EFFECTS

Manipulated through congruence or incongruence between n-1 and n trials

BASED ON EXPERIMENTAL BLOCKS USING EITHER THE ENDOGENOUS PROCEDURE (BUT WITH UNPREDICTIVE CUES) OR THE RHYTHMIC PROCEDURE (BUT ONLY WITH ASYNCHRONOUS RHYTHMS)

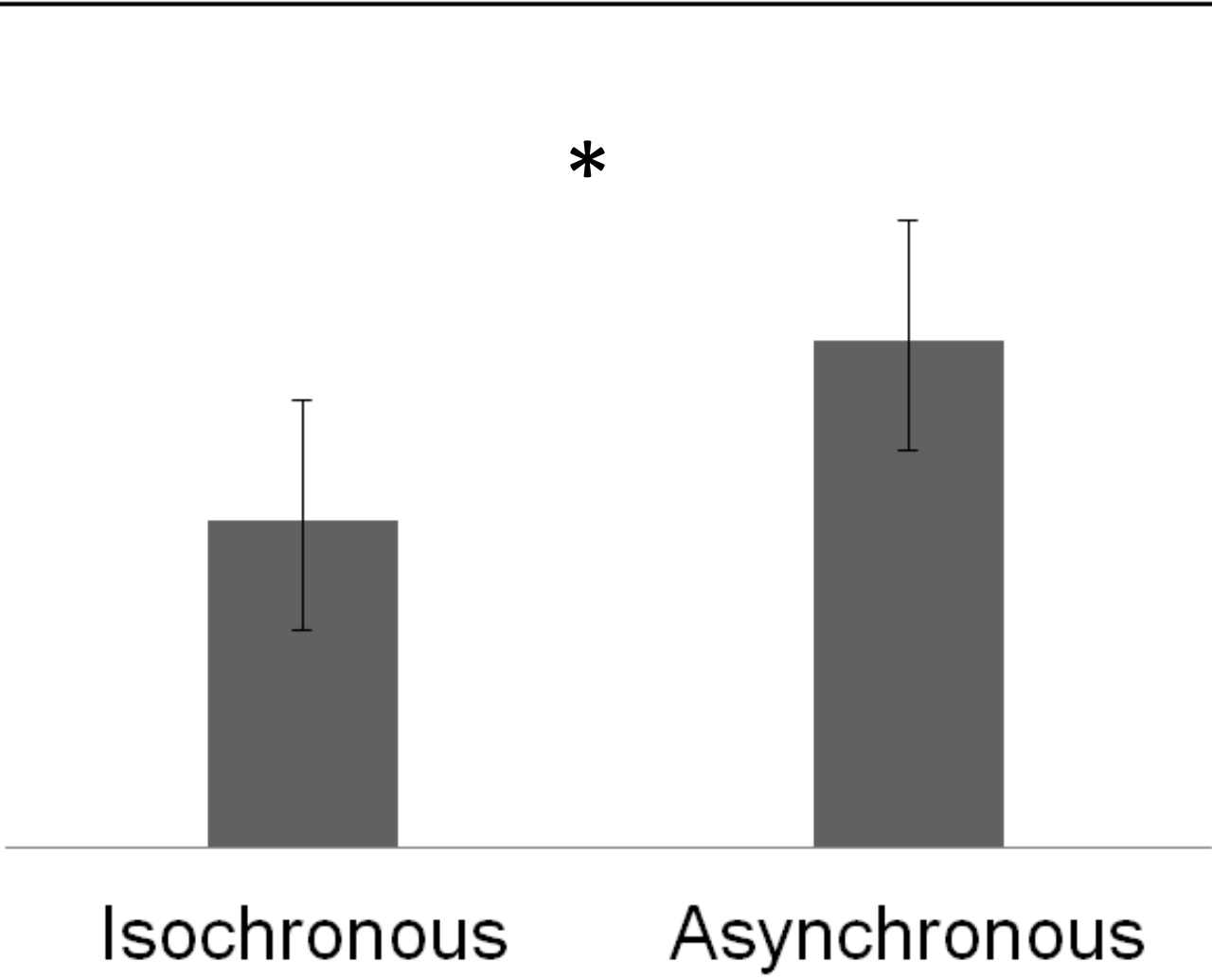
RESULTS

Endogenous orienting



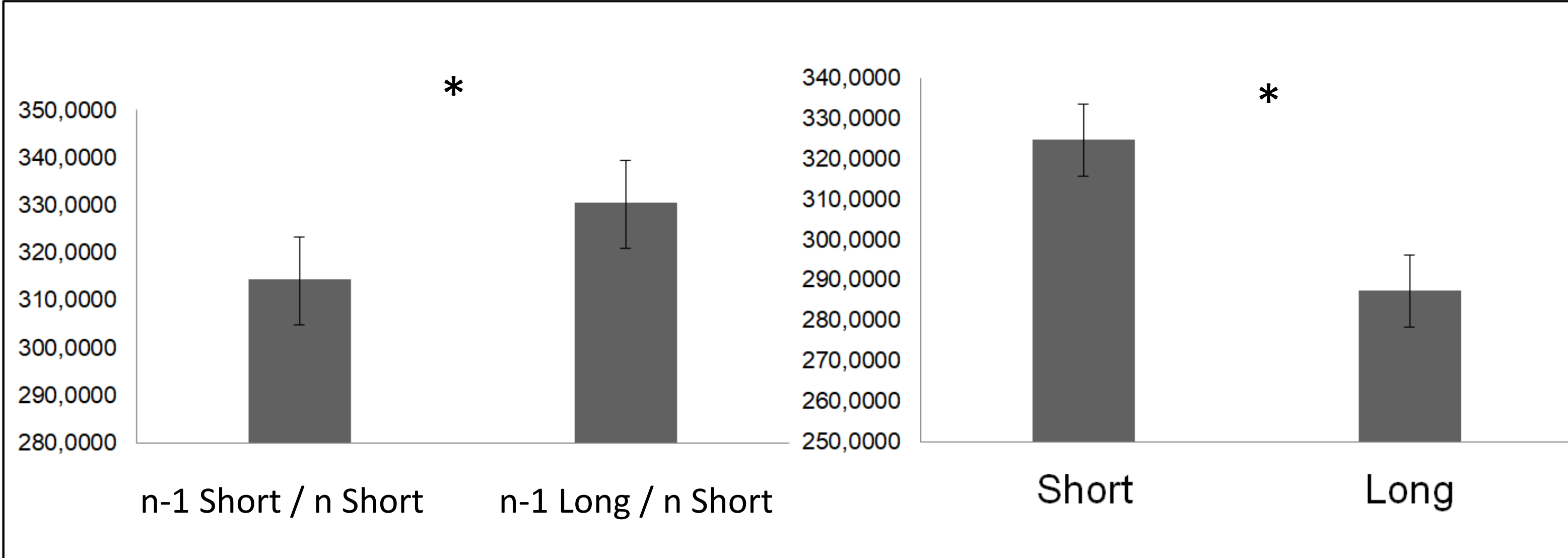
$F(1;23)=6.33$; $p<.05$

Rhythmic orienting



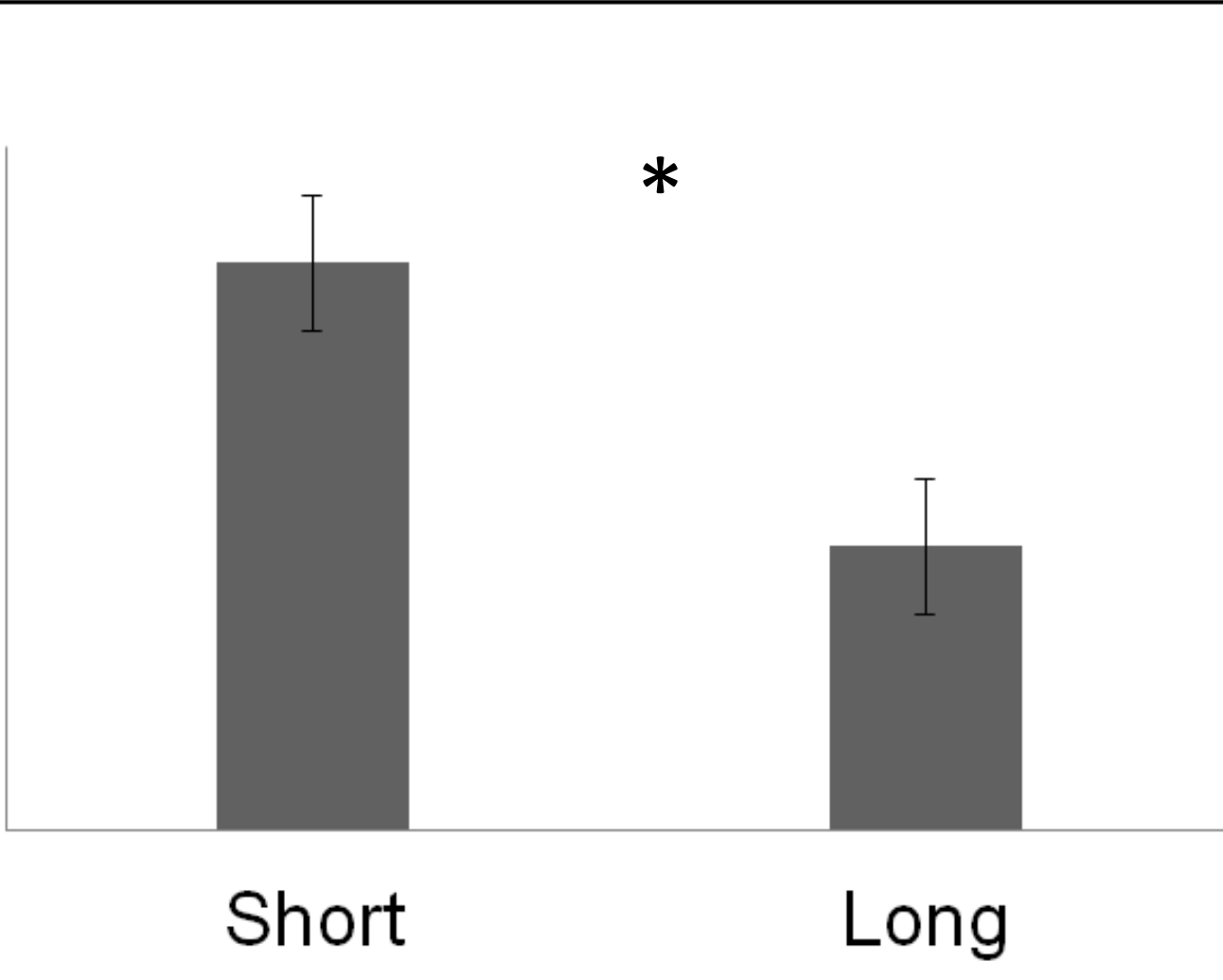
$F(1;23)=12.09$; $p<.01$

Sequential effects



$F(1;23)=34.88$; $p<.0001$

Foreperiod effect



$F(1;23)=23.05$; $p<.0001$

DISCUSSION

- This paradigm can be used among children to study developmental trajectories; with adults to investigate behavioural performance (correlations between the 4 components) but also to gain insight on pathologies encompassing temporal orienting deficits.
- Trivi o et al (2010) showed that deficits in some components of temporal preparation (endogenous temporal orienting and foreperiod effect) can be observed after prefrontal lesion; while other components are preserved. In other words, the 4 components of temporal preparation are likely to rely on distinct cognitive processes and cerebral bases. Our paradigm offers a straightforward experiment that provides “within” measures of these 4 components.

Follow-up studies

- This paradigm will be used to explore developmental trajectory in children aged 6 to 10.
- Indeed, automatic components of temporal preparation might emerge earlier than more endogenous modes of orienting (Droit-Volet & Coull, 2016; Coull & Droit-Volet, 2018)

Capizzi, M., & Correa,  . (2018). Measuring Temporal Preparation. In *Timing and Time Perception: Procedures, Measures, & Applications* (pp. 216–232). Brill.

Coull, J. T., & Droit-Volet, S. (2018). Explicit understanding of duration develops implicitly through action. *Trends in cognitive sciences*, 22(10), 923–937

Coull, J. T., & Nobre, A. C. (1998). Where and When to Pay Attention: The Neural Systems for Directing Attention to Spatial Locations and to Time Intervals as Revealed by Both PET and fMRI. *The Journal of Neuroscience*, 18(18), 7426–7435.

Droit-Volet, S., & Coull, J. T. (2016). Distinct developmental trajectories for explicit and implicit timing. *Journal of Experimental Child Psychology*, 150, 141–154

Cutanda D., Correa A., Sanabria D. (2015). Auditory temporal preparation induced by rhythmic cues during concurrent auditory working memory tasks. *Journal of Experimental Psychology : Human Perception and Performance*, 41, 790–797.

Nobre, A. C., & Van Ede, F. (2018). Anticipated moments: temporal structure in attention. *Nature Reviews Neuroscience*, 19(1), 34.

Sanabria, D., Capizzi, M., & Correa, A. (2011). Rhythms that speed you up. *Journal of Experimental Psychology. Human Perception and Performance*, 37(1), 236–244.

Trivi o M., Correa A., Arnedo M., Lupi   ez J. (2010). Temporal orienting deficit after prefrontal damage, *Brain*, 133, 1173–1185.